Lovely Professional University

INT234 - Predictive Analytics (CA-2)

PROJECT REPORT

Student Academic Performance Prediction Using Machine Learning

A comparative study of five classification models
(Gaussian Naive Bayes, Logistic Regression, Decision Tree, Random Forest and KNN)

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Course: INT234 - Predictive Analytics (CA-2)
Dataset: UCI Student Performance (student-mat.csv, Mathematics course)
Tools: Python, pandas, scikit-learn, matplotlib, seaborn

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Appendix A: How to Run the Project

Abstract

Predicting which students are at risk of failing a course helps teachers to act early. This project uses the UCI *Student Performance* dataset (Mathematics course, 395 students, 33 columns) to build a binary classifier that predicts whether a student will PASS (final grade G3 >= 10) or FAIL (G3 < 10). The final grade G3 was used only to create the target and was removed from the input features to prevent data leakage. After preprocessing (scaling of numeric and one-hot encoding of categorical variables) and a stratified 80/20 train-test split (316 training and 79 test students), five models were trained and compared: Gaussian Naive Bayes, Logistic Regression, Decision Tree, Random Forest and K-Nearest Neighbors.

On the held-out test set the **Decision Tree** obtained the highest F1-score (0.909) with accuracy 88.61%. Logistic Regression, Decision Tree and Random Forest all reached 87-89% accuracy, while Gaussian Naive Bayes (74.68%) and KNN (77.22%) were weaker. Because the test set is small, 5-fold stratified cross-validation was also performed; in cross-validation **Logistic Regression** was the most accurate model (92.15%). The earlier period grades G1 and G2 were by far the strongest predictors. An additional experiment without G1 and G2 reduced accuracy to about 58-67%, showing that predicting early, before any grades exist, is much harder.

Keywords: student performance, classification, data leakage, Logistic Regression, Random Forest, educational data mining.

1. Introduction

Education is one of the areas where data is collected every day: attendance, marks, family background and study habits. Educational Data Mining (EDM) applies machine learning to this data to understand and improve learning. One practical use is an *early-warning system* which flags students who are likely to fail so that teachers, mentors and parents can help them in time.

Machine learning classification algorithms learn patterns from past students (inputs such as age, study time, past failures, absences and earlier grades) and use these patterns to predict the result of a new student. In this project five well-known classifiers are applied to the Mathematics-course part of the Student Performance dataset collected from two secondary schools in Portugal (Cortez and Silva, 2008) and their performance is compared using accuracy, precision, recall, F1-score, ROC-AUC and confusion matrices.

2. Problem Statement

Given the demographic, social and school-related information of a secondary-school student together with the first- and second-period grades (G1, G2), predict whether the student will **PASS** (final grade G3 >= 10) or **FAIL** (G3 < 10) the Mathematics course. The task is a **binary classification** problem. The goal is to find which of the five selected models predicts the result most reliably and which factors influence the result most.

3. Objectives

- To load and inspect the student-mat.csv dataset (semicolon separated) and verify its columns.
- To convert the final grade G3 into a PASS/FAIL target and remove G3 from the inputs to avoid data leakage.
- To perform exploratory data analysis (EDA) and find the factors related to passing.
- To preprocess the data and create a stratified train-test split.
- To train five classifiers: Gaussian Naive Bayes, Logistic Regression, Decision Tree, Random Forest and KNN.
- To evaluate them with accuracy, precision, recall, F1-score, ROC-AUC, cross-validation and confusion matrices.
- To compare the models honestly and to identify the most important predictors.
- To test how much the prediction depends on the earlier grades G1 and G2.

4. Dataset Description

The dataset *student-mat.csv* belongs to the UCI Machine Learning Repository (Student Performance data set). It contains **395 students** and **33 columns** (30 attributes about the student, plus three grades G1, G2, G3). The file uses a semicolon (;) as separator, so it was read with `pd.read_csv(..., sep=';')`. After loading, the column names were checked against the attribute list in *student.txt* and all 33 columns matched. The data has **0 missing values** and **0 duplicate rows**, so no cleaning of missing data was required. Student ages range from 15 to 22 years.

Attribute	Type	Description
school, sex, address, famsize, Pstatus	Binary	School (GP/MS), sex, urban/rural, family size, parents living together/apart
age	Numeric	Student age (15 to 22)
Medu, Fedu	Ordinal 0-4	Mother's / father's education level
Mjob, Fjob, reason, guardian	Nominal	Parents' jobs, reason for choosing the school, guardian
traveltime, studytime	Ordinal 1-4	Travel time to school, weekly study time
failures	Numeric	Number of past class failures
schoolsup, famsup, paid, activities, nursery, higher, internet, romantic	Binary yes/no	Extra support, paid classes, activities, nursery, wants higher education, internet, relationship
famrel, freetime, goout, Dalc, Walc, health	Ordinal 1-5	Family relations, free time, going out, workday/weekend alcohol, health
absences	Numeric	Number of school absences (0 to 75 in this file)
G1, G2	Numeric 0-20	First and second period grades
G3	Numeric 0-20	Final grade - used ONLY to build the target (then removed)

Table 1: Attributes of the dataset (from student.txt)

Target variable. A new variable was created from G3: PASS if G3 >= 10, otherwise FAIL. This gives **265 PASS** (67.1%) and **130 FAIL** (32.9%) students, so the classes are moderately imbalanced. For this reason a *stratified* split and the F1-score are used. PASS is treated as the positive class when calculating precision, recall and F1.

Note on zero grades. 38 students have G3 = 0 (most likely students who did not sit the final evaluation). They were kept in the data, because they are real FAIL cases that an early-warning system should detect.

5. Methodology

The workflow of the project is shown below. Every step is implemented in the file `student_performance_prediction.py`.

Step	Action
1	Load CSV with sep=';' and inspect shape, columns, types, missing values and duplicates
2	Create target: PASS if G3 >= 10 else FAIL; drop G3 from features
3	Exploratory data analysis (distributions, correlations, pass rates, boxplots)
4	Stratified 80/20 train-test split (random_state = 42)
5	Preprocessing inside a Pipeline: StandardScaler (numeric) + OneHotEncoder (categorical)
6	Train five models on the training set
7	Evaluate on the unseen test set + 5-fold stratified cross-validation
8	Compare models, study feature importance, run extra experiment without G1/G2

Table 2: Project workflow

5.1 Data Leakage and Removal of G3

Data leakage happens when the input features contain information that would not be available at prediction time or that directly reveals the answer. Here the target (PASS/FAIL) is calculated directly from G3. If G3 stayed in the input, a model could learn the trivial rule 'G3 >= 10 means PASS' and show almost 100% accuracy, which would be meaningless. Therefore G3 was **removed from the feature set** (the code even checks `'G3' in X.columns` and prints False). The models use the remaining **32 features** (15 numeric and 17 categorical).

About G1 and G2. G1 and G2 are the grades of the first two periods. They are available before the final exam, so they are legitimate inputs for a mid-year prediction, and the assignment only asks to remove G3. However they are strongly related to G3 (see EDA) and dominate the predictions. For transparency, Section 9 repeats the experiment without G1 and G2.

5.2 Preprocessing

- **Numeric features** (15): standardised with StandardScaler (mean 0, standard deviation 1). This is important for KNN and Logistic Regression, which depend on distances / coefficient size.
- **Categorical features** (17): converted with OneHotEncoder, giving 58 input columns in total.
- The preprocessing is placed **inside a scikit-learn Pipeline**, so the scaler and encoder are fitted only on training data (inside cross-validation, only on the training folds). This avoids a second kind of leakage from the test data.
- No rows were removed: there are no missing values or duplicates.

5.3 Stratified Train-Test Split

The data was split 80% / 20% using `train_test_split(..., stratify=y, random_state=42)`. Stratification keeps the PASS/FAIL ratio equal in both parts: the training set has 316 students (212 PASS) and the test set has 79 students (53 PASS, 26 FAIL); both have 67.09% PASS.

5.4 Evaluation Metrics

- **Accuracy** = correct predictions / all predictions.
- **Precision** = TP / (TP + FP): of the students predicted PASS, how many really passed.
- **Recall** = TP / (TP + FN): of the students who really passed, how many were found.
- **F1-score** = harmonic mean of precision and recall.
- **Confusion matrix** = counts of TN, FP, FN, TP (rows = actual, columns = predicted).
- **ROC-AUC** = ability to separate PASS from FAIL at all thresholds (1.0 = perfect, 0.5 = random).
- **5-fold stratified cross-validation** on the full data, because the test set has only 79 students.

6. Exploratory Data Analysis

EDA was performed with pandas, matplotlib and seaborn. All graphs are saved as PNG files in the *outputs* folder. (G3 is used in EDA only to understand the data; it is not a model input.)

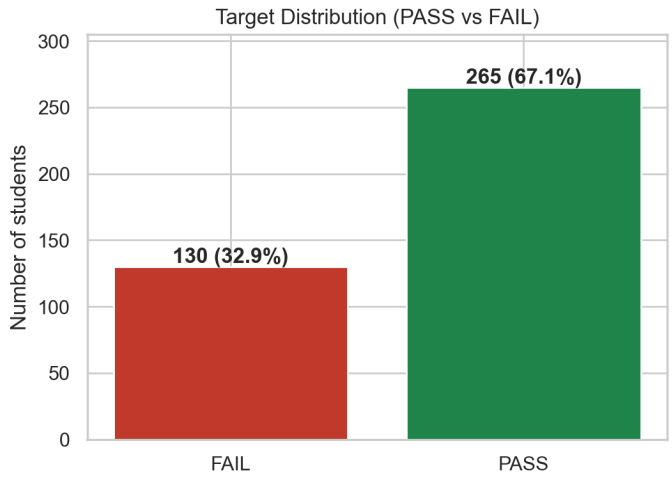


Figure 1: Distribution of PASS and FAIL students

The dataset is moderately imbalanced: 265 students passed and 130 failed. The mean final grade is 10.42 (median 11, standard deviation 4.58).

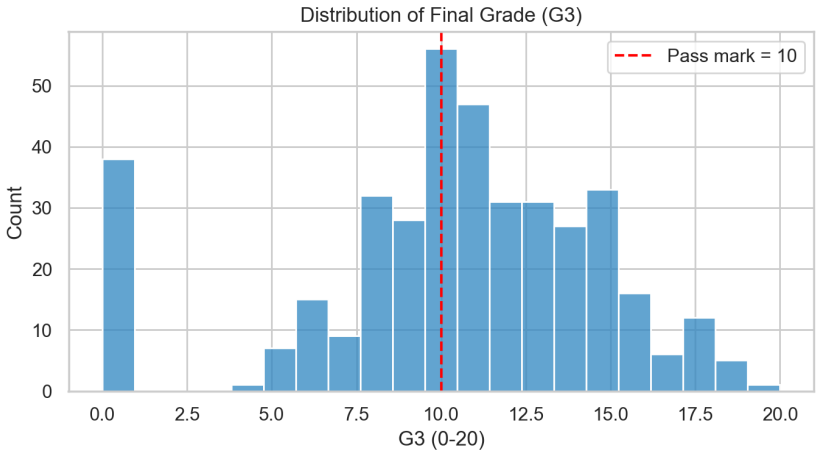


Figure 2: Histogram of the final grade G3 with the pass mark at 10

The histogram shows a cluster of students near grades 10-14 and a separate spike at 0 (38 students), which probably correspond to students who did not complete the final evaluation (this is an assumption; the dataset does not say).

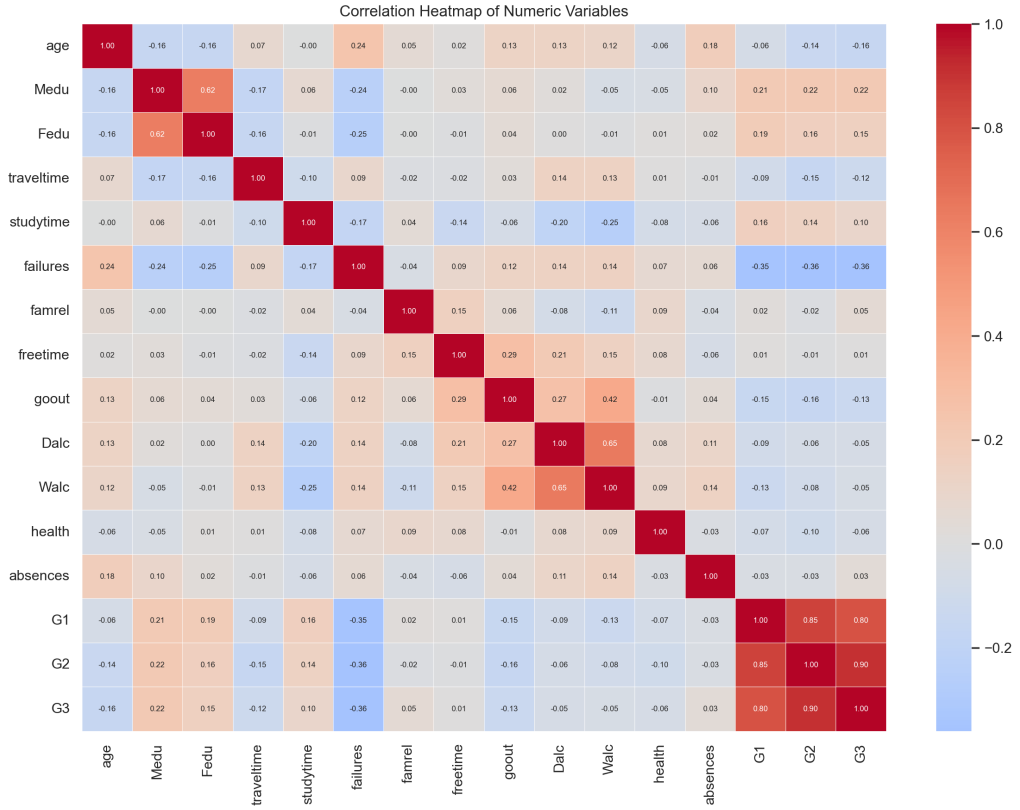


Figure 3: Correlation heatmap of numeric variables

The strongest correlations with G3 are G2 (+0.90) and G1 (+0.80), followed by past failures (-0.36), mother's education Medu (+0.22) and age (-0.16). Other variables have weak correlation (below 0.2 in absolute value).

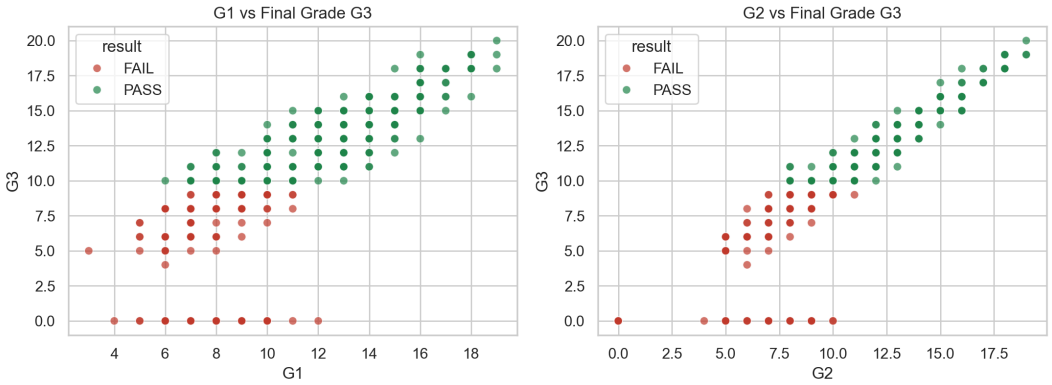


Figure 4: G1 and G2 compared with the final grade G3

The scatter plots show an almost straight-line relationship: students who score well in the first two periods usually also score well in the final period. This explains why G1 and G2 are so important for prediction.

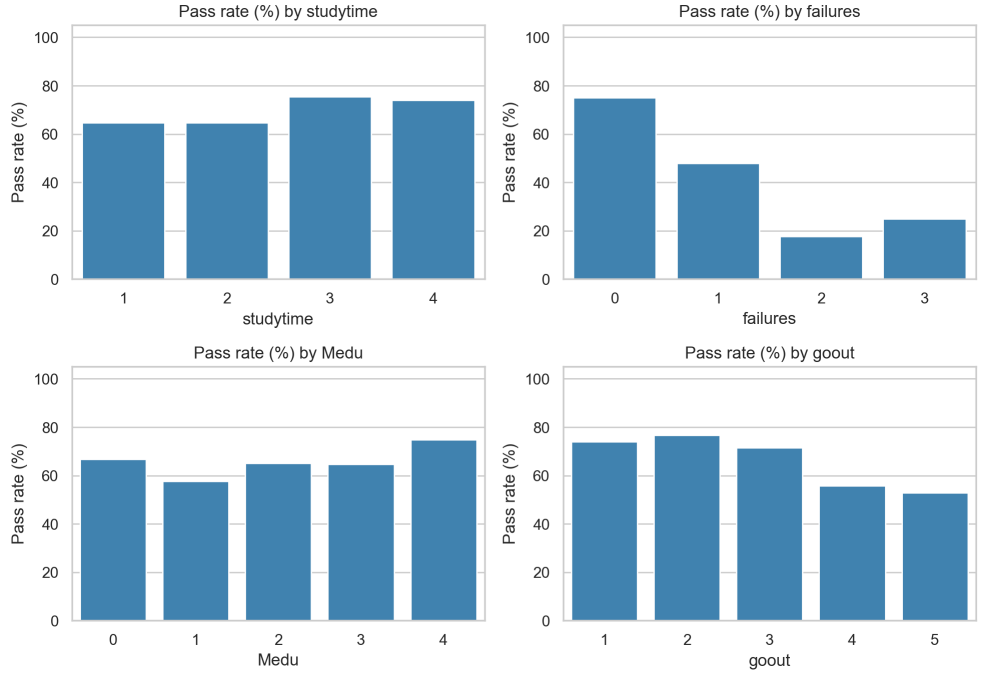


Figure 5: Pass rate (%) by study time, past failures, mother's education and going out

Past failures: pass rate falls from 75.0% (0 failures, n=312) to 48.0% (1 failure, n=50) and only 17.6% (2 failures, n=17). **Going out:** students who go out very often (levels 4 and 5) have lower pass rates (55.8% and 52.8%) than those who go out rarely (73.9% at level 1). **Mother's education:** pass rate is 57.6% when the mother has primary education only and 74.8% when she has higher education. **Study time:** pass rate is 64.8% for the lowest study-time group and 75.4% for level 3; the relationship is weak and not strictly increasing.

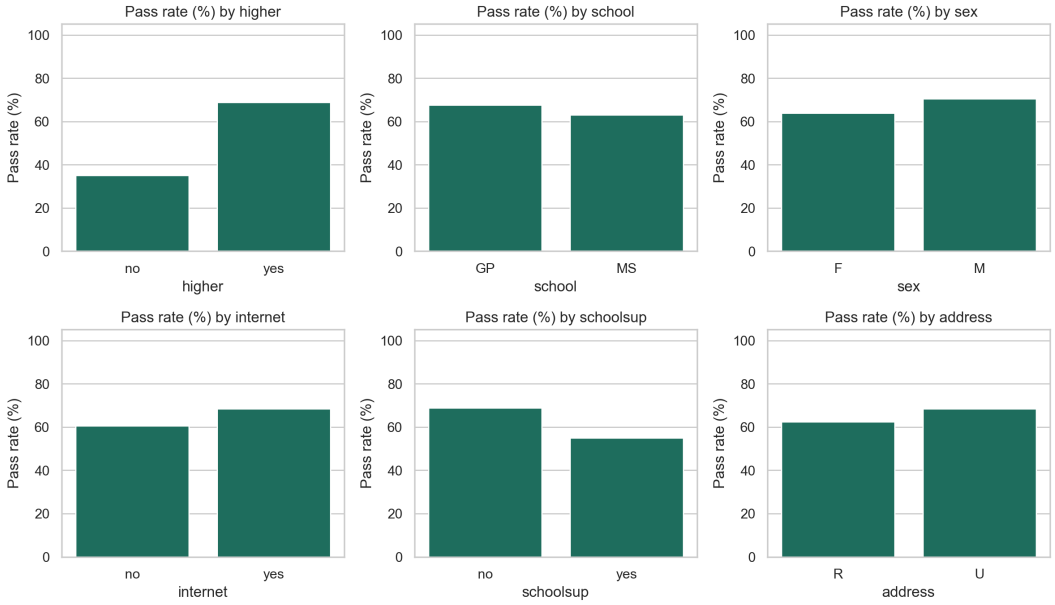


Figure 6: Pass rate (%) for categorical features

Students who want to take higher education pass more often (68.8% vs 35.0%; note the 'no' group has only 20 students). Students receiving extra school support have a lower pass rate (54.9% vs 68.9%), which probably means that support is given to students who are already weak, so this must not be read as 'support harms results'.

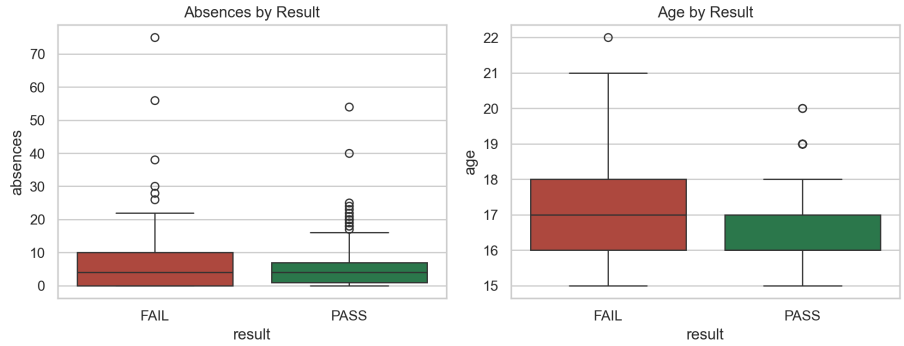


Figure 7: Absences and age by result

Failing students are slightly older on average (17.02 vs 16.54 years) and have more past failures on average (0.69 vs 0.16). Their mean G1 (7.76) and G2 (6.82) are far below those of passing students (12.45 and 12.62). Mean absences are 6.76 for FAIL and 5.19 for PASS; absences contain some large outliers (maximum 75).

7. Machine Learning Models

All five models were trained on the same training data and tested on the same test set (79 students). No extensive hyper-parameter tuning was done; sensible settings were chosen in advance so that the comparison is simple and not tuned on the test set. In the confusion matrices below, rows are the actual class and columns are the predicted class.

7.1 Gaussian Naive Bayes

Gaussian Naive Bayes applies Bayes' theorem and assumes that all features are independent of each other given the class, and that each numeric feature follows a normal distribution within each class. It is very fast and works with little data, but the independence assumption is not true here (for example G1 and G2 are strongly correlated, and one-hot columns are not Gaussian).

Settings used: `GaussianNB() with default settings.`

Accuracy	Precision	Recall	F1-score	ROC-AUC	CV accuracy (5-fold)
0.7468	0.8000	0.8302	0.8148	0.8498	0.8000 +/- 0.0218

Table 3: Gaussian Naive Bayes - test-set results (positive class = PASS)

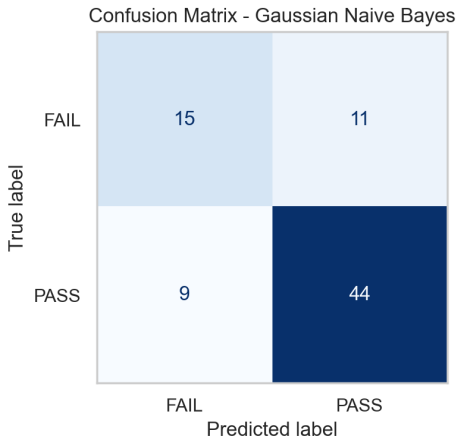


Figure 8: Confusion matrix of Gaussian Naive Bayes

Out of 79 test students, the model correctly identified **15** FAIL students (TN) and **44** PASS students (TP). It wrongly predicted PASS for **11** students who actually failed (FP) and wrongly predicted FAIL for **9** students who actually passed (FN). The training accuracy was 0.8513, compared with a test accuracy of 0.7468.

7.2 Logistic Regression

Logistic Regression models the probability of PASS as a sigmoid function of a weighted sum of the features. It is a linear, easily explainable model: the sign and size of each coefficient show the direction and strength of the effect (for standardised inputs).

Settings used: `LogisticRegression(max_iter=2000, random_state=42) with default L2 regularisation.`

Accuracy	Precision	Recall	F1-score	ROC-AUC	CV accuracy (5-fold)
0.8734	0.9574	0.8491	0.9000	0.9485	0.9215 +/- 0.0095

Table 4: Logistic Regression - test-set results (positive class = PASS)

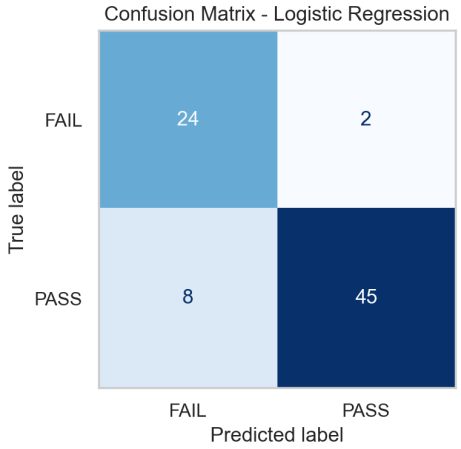


Figure 9: Confusion matrix of Logistic Regression

Out of 79 test students, the model correctly identified **24** FAIL students (TN) and **45** PASS students (TP). It wrongly predicted PASS for **2** students who actually failed (FP) and wrongly predicted FAIL for **8** students who actually passed (FN). The training accuracy was 0.9684, compared with a test accuracy of 0.8734.

7.3 Decision Tree

A Decision Tree repeatedly splits the data using the question that best separates PASS from FAIL (Gini impurity). The result is a set of simple if-then rules that are easy to explain. Unlimited trees overfit, so the depth was restricted.

Settings used: `DecisionTreeClassifier(max_depth=5, min_samples_leaf=5, random_state=42).`

Accuracy	Precision	Recall	F1-score	ROC-AUC	CV accuracy (5-fold)
0.8861	0.9783	0.8491	0.9091	0.9144	0.9139 +/- 0.0095

Table 5: Decision Tree - test-set results (positive class = PASS)

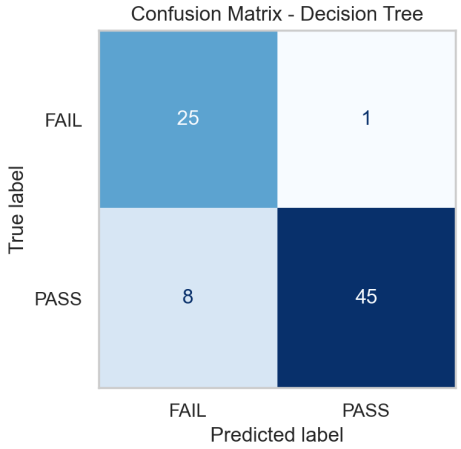


Figure 10: Confusion matrix of Decision Tree

Out of 79 test students, the model correctly identified **25** FAIL students (TN) and **45** PASS students (TP). It wrongly predicted PASS for **1** students who actually failed (FP) and wrongly predicted FAIL for **8** students who actually passed (FN). The training accuracy was 0.9557, compared with a

test accuracy of 0.8861.

7.4 Random Forest

Random Forest builds many decision trees on random samples of the data and random subsets of the features and combines their votes. This reduces overfitting compared with a single tree and provides feature importance values (Breiman, 2001).

Settings used: `RandomForestClassifier(n_estimators=300, random_state=42).`

Accuracy	Precision	Recall	F1-score	ROC-AUC	CV accuracy (5-fold)
0.8734	0.9574	0.8491	0.9000	0.9376	0.9063 +/- 0.0261

Table 6: Random Forest - test-set results (positive class = PASS)

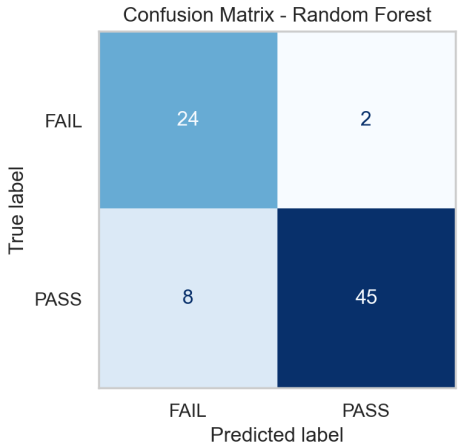


Figure 11: Confusion matrix of Random Forest

Out of 79 test students, the model correctly identified **24** FAIL students (TN) and **45** PASS students (TP). It wrongly predicted PASS for **2** students who actually failed (FP) and wrongly predicted FAIL for **8** students who actually passed (FN). The training accuracy was 1.0000, compared with a test accuracy of 0.8734.

7.5 K-Nearest Neighbors

KNN classifies a student by looking at the K most similar students in the training data (by Euclidean distance) and taking the majority class. It needs scaled features and can suffer when there are many irrelevant features, as is the case here (50+ columns after encoding).

Settings used: `KNeighborsClassifier(n_neighbors=5) on scaled and encoded data.`

Accuracy	Precision	Recall	F1-score	ROC-AUC	CV accuracy (5-fold)
0.7722	0.8182	0.8491	0.8333	0.8226	0.7975 +/- 0.0226

Table 7: K-Nearest Neighbors - test-set results (positive class = PASS)

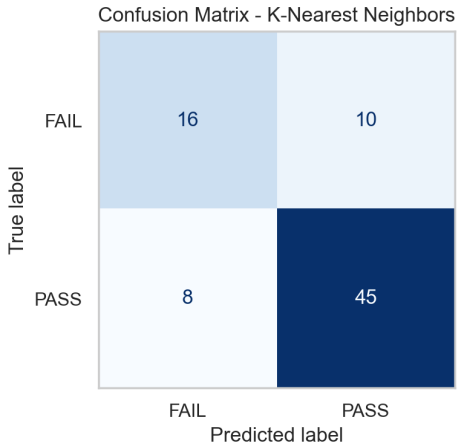


Figure 12: Confusion matrix of K-Nearest Neighbors

Out of 79 test students, the model correctly identified **16** FAIL students (TN) and **45** PASS students (TP). It wrongly predicted PASS for **10** students who actually failed (FP) and wrongly predicted FAIL for **8** students who actually passed (FN). The training accuracy was 0.8608, compared with a test accuracy of 0.7722.

8. Results and Model Comparison

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	CV Acc.	CV F1
Gaussian Naive Bayes	0.7468	0.8000	0.8302	0.8148	0.8498	0.8000	0.8526
Logistic Regression	0.8734	0.9574	0.8491	0.9000	0.9485	0.9215	0.9414
Decision Tree	0.8861	0.9783	0.8491	0.9091	0.9144	0.9139	0.9343
Random Forest	0.8734	0.9574	0.8491	0.9000	0.9376	0.9063	0.9297
K-Nearest Neighbors	0.7722	0.8182	0.8491	0.8333	0.8226	0.7975	0.8582

Table 8: Comparison of all five models (test set metrics; CV = 5-fold stratified cross-validation on all 395 students)

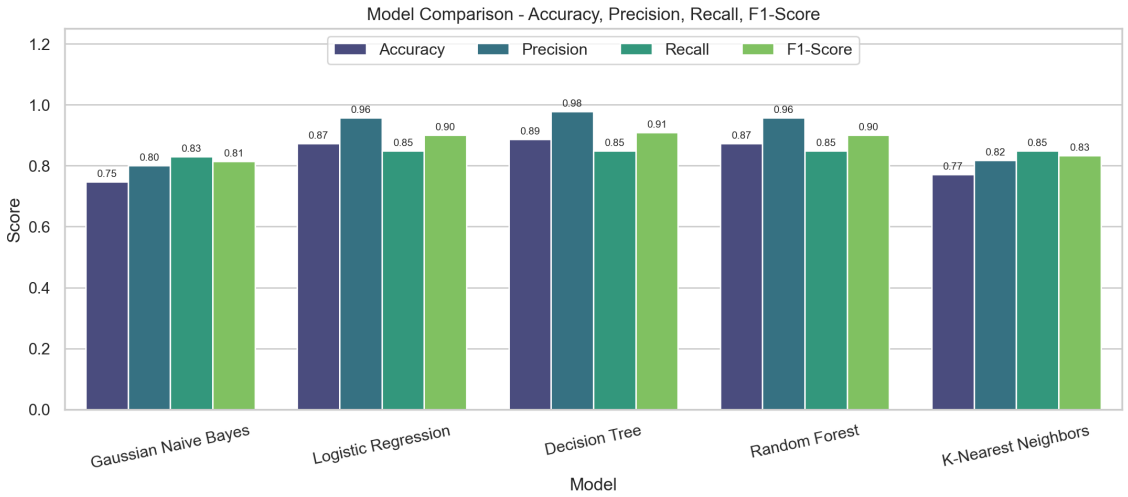


Figure 13: Accuracy, precision, recall and F1-score of the five models

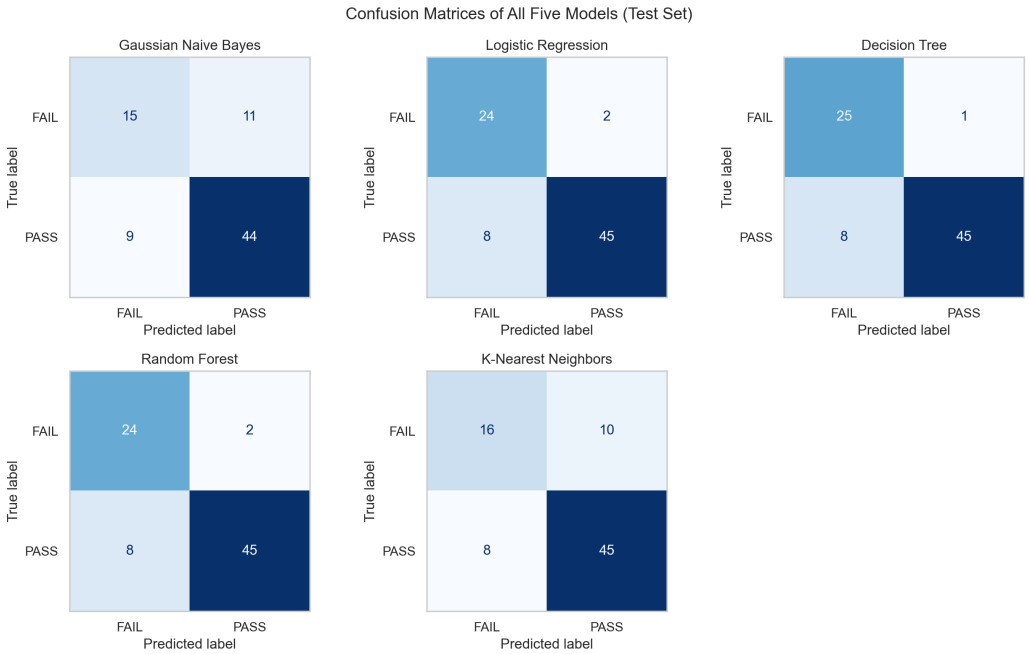


Figure 14: Confusion matrices of all five models

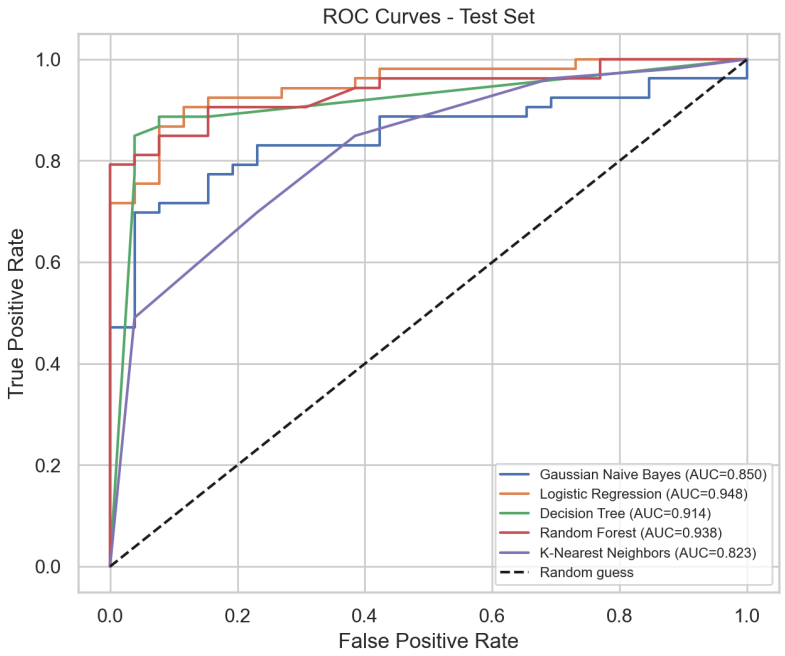


Figure 15: ROC curves of the five models on the test set

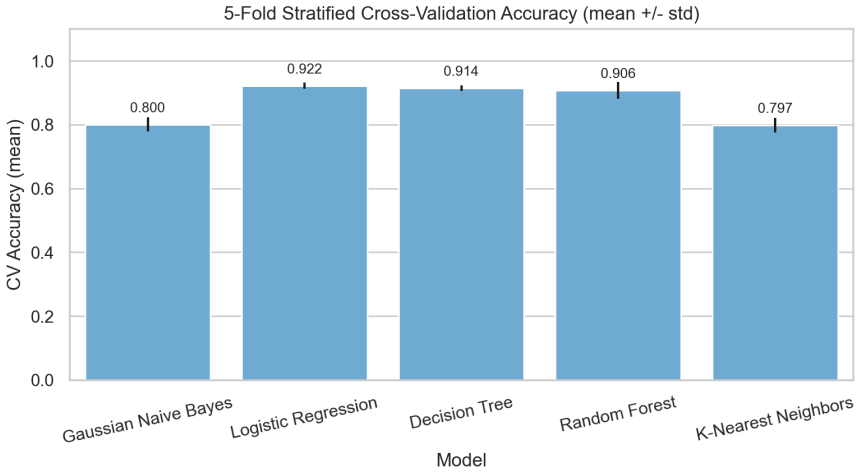


Figure 16: 5-fold cross-validation accuracy (mean and standard deviation)

8.1 Interpretation of the results

- **Best on the single test split:** Decision Tree (accuracy 0.8861, F1 0.9091). Logistic Regression and Random Forest are very close (accuracy 0.8734, F1 0.9000).
- **Test set is small:** one student equals 1.27 percentage points of accuracy, so the difference between Decision Tree (88.61%) and Logistic Regression / Random Forest (87.34%) is only **one student**. These three models should be regarded as roughly equal on this split.
- **Best in cross-validation:** Logistic Regression (92.15% accuracy, F1 0.9414), followed closely by Decision Tree (91.39%) and Random Forest (90.63%). Cross-validation uses all 395 students and is more reliable than a single split.
- **Weakest models:** Gaussian Naive Bayes (F1 0.815) and KNN (F1 0.833). Both made 10-11 errors on FAIL students (they predicted PASS for about 40% of the real FAIL students). The independence assumption (Naive Bayes) and the many irrelevant one-hot features (KNN) are likely reasons - this is an explanation, not something tested directly in this project.
- **Precision vs recall:** The three best models have very high precision for PASS (0.96-0.98) but lower recall (0.85): they rarely wrongly promise a pass, but miss about 8 students who pass. For FAIL students they find 24-25 out of 26 (recall 0.92-0.96), which is useful for an early-warning system.
- **Overfitting signs:** Random Forest had training accuracy 1.00 versus test accuracy 0.87, and Logistic Regression had 0.97 vs 0.87. Some gap is normal, but it shows that the models fit the training data more closely than new data.

8.2 Important features

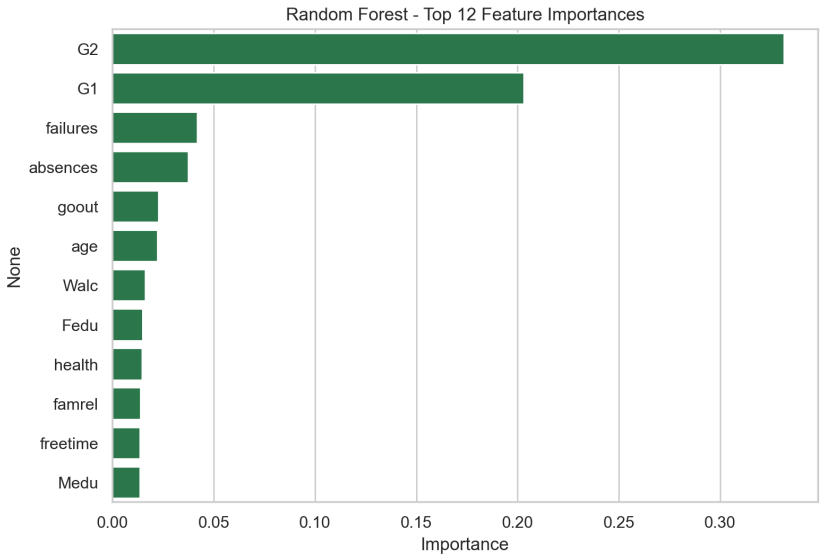


Figure 17: Top 12 Random Forest feature importances

In the Random Forest, G2 (0.332) and G1 (0.203) together account for more than half of the total importance, followed by past failures (0.042) and absences (0.037). In the Logistic Regression the largest coefficients (standardised inputs) are G2 (+3.98) and G1 (+1.82). The Decision Tree relies almost entirely on G2 (importance 0.85).

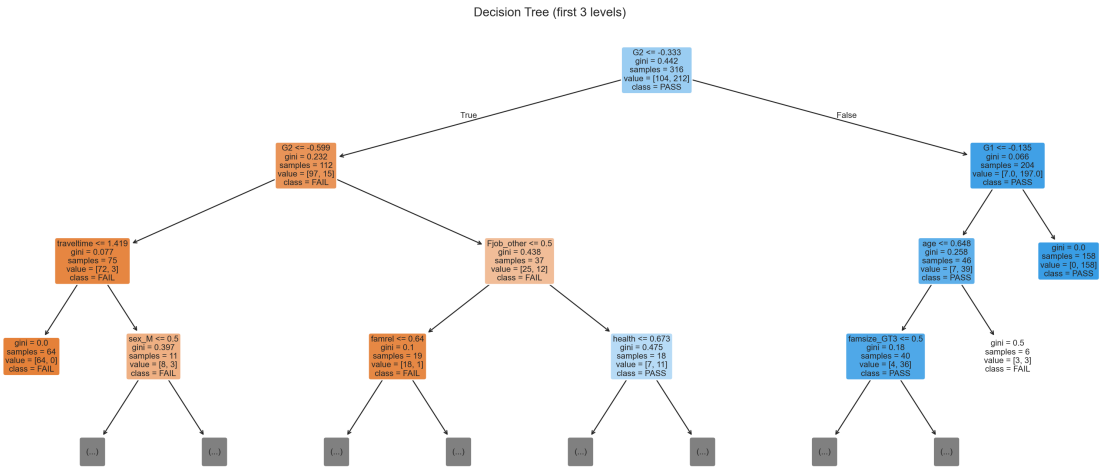


Figure 18: Decision Tree (first three levels)

9. Additional Experiment: Predicting Without G1 and G2

Because G1 and G2 dominate the prediction, the whole experiment was repeated using the same split and the same model settings, but with G1 and G2 also removed. This simulates predicting *before* any grade of the course is available.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	CV Acc.
Gaussian Naive Bayes	0.6456	0.7049	0.8113	0.7544	0.6089	0.6633
Logistic Regression	0.6582	0.7167	0.8113	0.7611	0.5885	0.6785
Decision Tree	0.6456	0.6984	0.8302	0.7586	0.6390	0.6759
Random Forest	0.6709	0.7077	0.8679	0.7797	0.6393	0.6911
K-Nearest Neighbors	0.5823	0.6613	0.7736	0.7130	0.5221	0.6354

Table 9: Results without G1 and G2 (test set; CV = 5-fold cross-validation)



Figure 19: F1-score with and without previous grades

Without G1 and G2 the test accuracy of all models falls to between 58.2% and 67.1% and the ROC-AUC to between 0.52 and 0.64, which is only slightly better than random guessing (0.50) for several models. Note that always predicting PASS would already give 67.1% accuracy, so models in this setting add little value. The F1-score of the PASS class remains around 0.71-0.78 mainly because the models predict PASS for most students. **Conclusion of this experiment:** social and family variables alone are weak predictors; the previous grades carry almost all of the predictive information.

10. Discussion

- The high scores in Section 8 are mainly due to G1 and G2. This is legitimate for a mid-year warning system but should not be mistaken for the ability to predict from background information alone.
- Simple models (Logistic Regression, shallow Decision Tree) performed as well as the Random Forest, which suggests that the relationship between the earlier grades and the final result is almost linear.
- Past failures, absences, age and going out were the main non-grade factors, which agrees with the EDA, but their effect is small compared with the grades.

- Teachers can use the models' high recall for FAIL students (up to 96%) to select students for extra help; the cost is a few false alarms.

11. Limitations

- Small dataset (395 students) and small test set (79 students); results can change with a different random split.
- Data comes from two schools in Portugal and one subject (Mathematics), so the results may not generalise to other schools or subjects.
- No extensive hyper-parameter tuning was done (by design, to keep the comparison simple).
- The pass mark of 10 converts a numeric grade into two classes and hides how close a student is to the mark.
- The relationship between support variables and results (for example extra school support) is correlation, not cause.

12. Conclusion and Future Scope

This project built and compared five classification models for predicting whether a student will pass the Mathematics course. G3 was used only to create the target and was removed from the inputs to avoid data leakage. On the held-out test set, the Decision Tree gave the best F1-score (0.909), with Logistic Regression and Random Forest practically equal; in 5-fold cross-validation Logistic Regression was the most accurate (92.15%). Gaussian Naive Bayes and KNN were clearly weaker. Since the three top models differ by only one test student, **Logistic Regression is recommended as the most reliable and most explainable model** for this data because it was the best in cross-validation and its coefficients are easy to explain. The earlier grades G1 and G2 were the strongest predictors; without them accuracy drops to the 58-67% range.

Future scope: tune hyper-parameters with GridSearchCV, try gradient-boosting models, combine the Mathematics and Portuguese datasets, predict the exact grade with regression, handle class imbalance (for example class weights), and build a simple web dashboard for teachers.

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Appendix A: How to Run the Project

- Install the libraries: `pip install pandas numpy matplotlib seaborn scikit-learn reportlab pillow`
- Put `student_performance_prediction.py` and `student-mat.csv` in the same folder.
- Open the .py file in Spyder and press **F5**. The folder `outputs/` is created with all graphs, CSV result tables, classification reports and results_summary.json.
- Run `generate_report.py` to rebuild this PDF from the saved results.
- Run `node build_ppt.js` (Node.js + pptxgenjs) to rebuild the PowerPoint.

All numbers in this report are read automatically from the files produced by the Python program, so the report always matches the code. Random seed 42 is used everywhere; results may differ slightly if a different scikit-learn version is installed.